

Abdelrahman Taeha

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science - Artificial Intelligence Concentration

Atlanta, GA
Aug. 2026 – May 2028 (Expected)

Sacramento State University

Bachelor of Science in Computer Science

Sacramento, CA
Aug. 2023 – Dec. 2025

EXPERIENCE

Software Engineer

DHCS

May 2025 – Present

Sacramento, CA

- Automated monitoring and reporting pipelines across enterprise systems, reducing manual observability overhead by 40% and cutting MTTD for production incidents in half.
- Built backend integrations and validation workflows that surfaced system anomalies early, accelerating root-cause analysis by 35% and streamlining incident response.
- Established internal dashboards visualizing agent-driven operational metrics in real time, reducing incident resolution time by 30% and improving real-time alerting accuracy by 20%.

AI Research Engineer

Medical AI Research

Aug. 2025 – Present

Remote

- Co-authored [SpineFairBench](#) (submitted NeurIPS 2026), a counterfactual benchmark evaluating 7,996 paired spinal radiographs to audit demographic sensitivity in VLM-based report generation.
- Evaluated medical LLM agent integrity across 840 paired-twin scenarios for the MedInsider FHIR-style benchmark, identifying systematic failure modes under simulated institutional pressure.
- Architected evaluation and logging infrastructure with deterministic action tracking and auditing workflows for 9 distinct medical vision-language models.

Software Engineering Intern (Machine Learning)

District Hut

May 2024 – Jul. 2025

Sacramento, CA

- Scaled distributed data pipelines processing 3 TB/week across cloud infrastructure, improving throughput and system reliability in a high-volume production environment.
- Revamped data integrity checks with SQL, preventing five critical production issues and automating 75% of data validation tasks with 100% test coverage.
- Collaborated with a team of 3 engineers to automate model retraining agents, reducing collective manual effort by 6 hours per week via autonomous performance metrics.

PROJECTS

R.A.D.A.R — Vision-Language Radiology Report Generator — github.com/abdeltachass/RADAR

- Built and deployed an end-to-end vision-language system that generates chest-X-ray reports from images, parameter-efficient fine-tuning BLIP-2 with LoRA (training 0.33% of 3.76B params) on 3,337 patient-split IU X-Ray studies.
- Quantified and mitigated hallucination (the core VLM failure mode) with reference-grounded analysis across decoding strategies, cutting the hallucination rate from 40% to 10% via confidence-threshold abstention.
- Engineered a clinical-term overlap metric exposing a fluent baseline scoring 44 ROUGE-L but 0 clinical-F1; shipped an interactive Gradio demo with per-sentence Grad-CAM grounding on a quantized Hugging Face Space.

S.A.I.N.T. — Network Intrusion Detection System — github.com/abdeltachass/S.A.I.N.T

- Partnered with 2 peers to build an AI agent reasoning engine for threat detection, reaching 89% accuracy on NSL-KDD (125K+ samples) via feature engineering across 41 traffic-flow features.
- Designed a low-latency inference agent using Flask + Redis + Plotly, processing 50–100 network connections/sec with an interactive agent-in-the-loop dashboard for analyst review.

PingVault — Network Observability Agent — github.com/abdeltachass/PingVault

- Created a production-grade network monitoring tool spanning 3 protocols (ICMP, SNMP, TCP/UDP) with agent-based metrics collection, Alembic migrations, and a real-time HTML observability dashboard mirroring enterprise SRE tooling.

TECHNICAL SKILLS

Data & ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, model evaluation, SAP (Data Integration)

AI & Agents: Autonomous Agents, LLM Orchestration (LangChain/CrewAI), RAG, Prompt Engineering

Programming: Python, Java, C/C++, SQL, JavaScript, Swift, MATLAB, R

Frameworks/Platforms: React.js, Node.js, Spring Boot, AWS, GCP, Docker, Git

Security/DevOps: SIEM, GitHub Actions, CI/CD, monitoring pipelines